

Hypothesis 3 - Effect of Positive Emotion on Working-Memory Performance

Confirmatory Hypothesis Report

Hypothesis

Positive emotional induction improves working-memory performance relative to neutral and negative emotional conditions.

Method

A confirmatory trial-level analysis was conducted to determine whether positive emotional induction improved working-memory performance relative to neutral and negative emotional conditions. The analytical sample included 23 participants.

The primary outcomes were response accuracy and reaction time. Accuracy was analyzed using generalized estimating equation (GEE) models with a binomial distribution, logit link, exchangeable within-participant correlation structure, and robust standard errors. Two planned contrasts were evaluated: Positive versus Neutral and Positive versus Negative. The accuracy analysis included 17,820 trial-level observations.

Reaction time was analyzed among valid correct-response trials only. Because reaction-time data were positively skewed, reaction time was log-transformed prior to modeling. Linear mixed-effects models were fitted with emotional condition as a fixed effect and a participant-specific random intercept. The reaction-time analysis included 14,184 observations.

GEE coefficients were exponentiated and interpreted as odds ratios. Log reaction-time coefficients were exponentiated and converted to percentage changes. Inverse efficiency scores were examined as a secondary indicator of combined speed and accuracy. Statistical significance was evaluated using a two-sided alpha level of .05, with 95% confidence intervals reported.

Results

Accuracy

Mean trial-level accuracy was 85.1% in the positive condition, 85.0% in the neutral condition, and 83.8% in the negative condition.

Positive emotion did not significantly improve accuracy relative to the neutral condition (OR = 1.002, 95% CI [0.90, 1.12], $p = .979$).

Accuracy was descriptively higher under positive than negative emotion, but this difference was not statistically significant (OR = 1.089, 95% CI [0.97, 1.22], $p = .137$).

Thus, the accuracy analyses did not provide evidence that positive emotional induction improved the probability of producing a correct response relative to either comparator.

Table 1. Descriptive accuracy across emotional conditions.

Condition	Participants	Trials	Correct	Accuracy	SD	Median
Negative	22	5940	4978	83.8%	0.368	1.0
Neutral	23	5940	5049	85.0%	0.357	1.0
Positive	22	5940	5055	85.1%	0.356	1.0

Table 2. Planned GEE contrasts for response accuracy.

Contrast	Beta	OR	95% CI	p
Positive vs Neutral	0.002	1.002	[0.90, 1.12]	= .979
Positive vs Negative	0.086	1.089	[0.97, 1.22]	= .137

Reaction Time

Positive emotion was associated with a substantial reduction in reaction time relative to the neutral condition.

Compared with neutral emotion, positive emotion was associated with a significantly lower log reaction time (beta = -0.285, SE = 0.015, $z = -18.60$, $p < .001$). After back-transformation, this corresponded to an estimated 24.8% reduction in reaction time.

By contrast, reaction time under positive emotion did not significantly differ from reaction time under negative emotion (beta = -0.020, SE = 0.016, $z = -1.31$, $p = .190$). The estimated difference was only 2.0%.

Therefore, positive emotional induction was associated with faster responding relative to the neutral condition, but it did not confer a significant reaction-time advantage over negative emotion.

Table 3. Descriptive reaction time across emotional conditions.

Condition	Participants	Trials	Mean RT	SD	Median RT	Mean log RT
Negative	22	4704	740.25	744.23	433.50	6.25
Neutral	23	4829	966.58	896.36	630.00	6.51
Positive	22	4651	771.33	808.86	421.00	6.24

Table 4. Planned mixed-effects contrasts for log reaction time.

Contrast	Beta	RT change	95% CI	p
Positive vs Neutral	-0.285	-24.8%	[-27.1%, -22.5%]	< .001
Positive vs Negative	-0.020	-2.0%	[-5.0%, 1.0%]	= .190

Processing Efficiency

Inverse efficiency was examined as a secondary measure of combined processing speed and accuracy.

For the Positive versus Neutral comparison, mean IES was 899.42 under positive emotion and 1126.67 under the neutral condition ($t = -4.38$, $p < .001$).

For the Positive versus Negative comparison, mean IES was 869.79 under positive emotion and 851.74 under negative emotion ($t = 0.21$, $p = .839$).

These secondary analyses should be interpreted alongside the separate accuracy and reaction-time models rather than as replacements for the primary outcomes.

Table 5. Paired inverse-efficiency comparisons.

Contrast	N	Positive IES	Comparator IES	Difference	t	p
Positive vs Neutral	22	899.42	1126.67	-227.25	-4.38	< .001
Positive vs Negative	21	869.79	851.74	18.05	0.21	= .839

Decision Rules

- H3 is supported if positive emotion improves performance relative to both neutral and negative conditions.
- H3 is partially supported if positive emotion improves only one primary outcome or improves performance relative to only one comparator.
- H3 is not supported if positive emotion produces no reliable performance advantage relative to either comparator.

Interpretation

Hypothesis 3 was partially supported.

Positive emotional induction did not improve response accuracy relative to either the neutral or negative condition.

However, participants responded substantially faster under positive emotion than under the neutral condition, with an estimated reaction-time reduction of 24.8%. This improvement in response speed occurred without a statistically detectable loss of accuracy.

The positive condition did not significantly outperform the negative condition in either accuracy or reaction time. Therefore, the observed facilitation cannot be interpreted as a valence-specific advantage of positive emotion.

Overall, positive emotional induction enhanced processing speed relative to the neutral baseline but did not produce a generalized improvement in working-memory performance across both comparator conditions.

Hypothesis Assessment

Conclusion: Hypothesis 3 was partially supported. Positive emotional induction significantly improved response speed relative to the neutral condition, but it did not improve accuracy and did not significantly outperform the negative condition.